

Optimizing Bank Marketing Campaigns through Machine Learning-Based Predictive Analytics

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ABSTRACT

In the modern world, banks rely heavily on marketing strategies and the engagement of customers for the growth of the business. This study analyzes the bank marketing dataset with the main aim of predicting whether the customers or clients will subscribe to a term deposit or not. The research focused on exploring the dataset, preprocessing, model training and making predictions about whether the customer will subscribe to the term deposit or not using R programming language. Important patterns were revealed through exploratory data analysis and visualization which offered insights into factors like age, job type and account balance of customers that could influence the clients in making decisions regarding term deposits. Subsequently, dataset went through several preprocessing steps to address missing values, handle outliers, normalization of numeric features and oversampling techniques were applied to handle class imbalance to ensure that the predictive modeling is based on reliable data. Three machine learning algorithms were used in this project which include Naïve Bayes, logistic regression and random forest. The performance of these three models was evaluated using evaluation metrics such as accuracy, precision, recall, F1 score, and the Area Under the ROC Curve (AUC). This evaluation of the models suggested that the Random Forest model outperformed the other two models, achieving an accuracy of 92.18%, precision of 88.09%, recall of 97.55%, F1-score of 92.58%, and AUC of 0.985. In comparison, Logistic Regression achieved 77.53% accuracy and Naïve Bayes achieved 77.36%. Additionally, the study demonstrates the ability of the selected models to predict the behavior of the customers and provides actionable insights for making crucial decisions in the banking sector to target the right customers with correct marketing strategies.

1 Introduction

In today's modern world, majority of businesses are using the latest and more advanced marketing strategies to attract new customers and retain them. The competition in the banking sector is increasing rapidly and the banking industry is looking for new innovative techniques to improve relationships with customers and engage more positively. Machine Learning (ML) is a branch of Artificial Intelligence (AI) which enables systems to learn from data and then make decisions without being explicitly programmed. Machine learning analysis large datasets and then uncover different patterns to predict the behavior of the customers. This helps banks to develop better marketing strategies and improve their customer services (Datacamp, 2025).

One important and crucial use of machine learning in the banking sector is for the development of marketing strategies and promoting the term deposit products. Old and traditional marketing strategies are not very effective now, therefore using machine learning models can result in better marketing campaigns, resulting in higher return on investment (ScienceDirect, 2025). However, according to McKinsey & Company (2025) the number of banks that have completely applied different machine learning models across their customer interactions is less than 10 percent. This clearly indicates that there is still a lot of room for improvement in the marketing strategies. This study focuses on exploring practical machine learning methods to address these limitations.

The main objective of this study is to explore how machine learning techniques can be applied in bank marketing to boost customer engagement which ultimately results in business growth. By careful data preprocessing, exploratory data analysis (EDA) and using advanced predictive techniques, the research aims to develop a clear process for using machine learning in banking.

The dataset used in this study is the Bank Marketing dataset, developed by Moro, Cortez, and Rita (2014). This dataset contains 45,211 records from a Portuguese bank's telemarketing campaigns. These marketing campaigns took place between 2008 and 2013. This dataset includes 17 variables which are related to customer demographics, financial data, and results of past marketing campaigns. One of the major issues with this dataset is that there are only about 11.3% of customers who subscribed to the term deposit . This imbalance in the dataset can result in difficulty while model training as most classifiers are more likely to favor the majority class (Kaggle, 2025).

Data preprocessing includes multiple steps. Missing values and outliers were handled carefully. To ensure the quality of the data, categorical entries that were marked as “unknown” were treated as missing values and then removed from the data. Similarly, outliers were handled using the interquartile range (IQR) method to avoid their negative impact on the results and performance of the model. Similarly, using min-max scaling data was normalized to bring values within a common range. The issue of class imbalance was solved by using a package called ROSE. Synthetic samples were generated for the minority class so that the model can learn from even distribution of the data.

Exploratory Data Analysis (EDA) was done for the better and thorough understanding of the dataset. Visualizations were created using tools such as ggplot2 which demonstrated different patterns and relationships between different features. Three machine learning algorithms were selected to predict whether the customer will subscribe to the term deposit or not. These ML models include Logistic Regression, Naïve Bayes and Random Forest. A package called caret was used for the fine tuning of the models. The model evaluation was done using different evaluation metrics such as F1-score, accuracy, area under the curve (AUC), precision and recall. This evaluation of the models suggested Random Forest model performed better than the other two models on the Bank Marketing dataset.

The following sections provide a detailed literature review, a detailed description of the methodology, present the results and findings and conclude the report with discussions, implications, and suggestions for future research.

2 Literature Review

Traditional marketing methods are no longer reliable due to increasing competition in the financial sector. The use of machine learning for developing marketing strategies in banks has increased in the past few years. One of the main reasons for this is the availability of data. As a lot of data about the customers is available, banks want to improve their marketing strategies to target the right customers. Machine learning helps to predict the behaviors and interests of the customers and thus improves the probability of getting positive responses from the clients or customers. One of the studies by Vashi et al. (2024) demonstrate that banks can benefit from data mining by making better decisions as it helps to predict how different customers will respond to different marketing strategies. According to this study, multiple factors like age, job type and amount of loan taken can have a huge influence on how a person can respond to different offers. Customers preference can be predicted by using data smartly.

Different machine learning techniques can be used as a combination to develop marketing strategies, and this approach has proven to be effective in most cases. One of the studies published in *PLOS ONE* (2023) suggested that ensemble learning techniques like random forest and XGboost performed better and gave better or more accurate results in predicting whether a customer will respond to a particular offer. The performance of the models enhanced when they were used as combination rather than using only one algorithm. This approach is more effective in situations where data is not balanced, for example in bank marketing datasets where most people would say no to subscribe to a term deposit (PLOS ONE, 2023).

Banks are not just limited to machine learning techniques to improve marketing strategies. They also use artificial intelligence tools like natural language processing to enhance their communication and engagement with the customers. One of the studies by Gerling and Lessmann (2024) explained how banks are using artificial intelligence and natural language processing for automation of customer service, understanding what interests the customer and personalizing messages. They found out that by using these technologies banks can find what customers like and thus they develop matching strategies. Using a combination of machine learning and natural language processing mostly for text-based data, banks improve their interactions with the customers.

Another study by Gao et al. (2024) provides a detailed explanation on the use of machine learning in banking and financial sectors. They focus on areas like the behavior of customers, fraud detection and management of risk involved in finance. They concluded that machine learning is now part of most of the big companies and is becoming a crucial part of banking and the financial sector. According to them, most used models include support vector machines, random forest and logistic regression. They highlighted that not only are these models useful, but they are also very easy to understand and implement. This study is therefore for researchers and people involved in the banking sector who think about the growth of business.

Even though machine learning has been implemented successfully in the banking sector, and it has helped banks to improve their marketing strategies. There are still a lot of challenges that need to be addressed. Some of these issues include quality of data, imbalanced dataset, and interpretation of models. For example, consider the data set of bank marketing, in this dataset majority of customers have refused to accept term deposit offer and thus it becomes difficult for the model to find patterns without being biased. This issue can be solved by using techniques such as oversampling and ensemble learning, however it can result in increased complexity of the model. Since the decisions made by these models can affect the customers, it is extremely important for the banks to make sure that the models are clear and easy to understand along with being able to accurately predict.

The literature review clearly indicates that banks use machine learning to enhance their marketing strategies. Different artificial intelligence and predictive analytics tools make it easy for banks to have a better understanding of the customers' behavior and target the right customers. However, it is extremely important for banks to do data cleaning, prepare data correctly, select the most appropriate models and test them regularly to make sure the use of models is fair and ethical.

3 Methodology

In this project, data driven method or technique was used to make predictions about the customers, whether they would agree to subscribe to the term deposit, using different supervised machine learning techniques. The entire process was broken down into simple steps that are used in predictive analytics. It starts with understanding the data, performing preprocessing steps like data cleaning, exploring multiple useful patterns, model building and then finally testing the model and evaluation of the results (James et al., 2013).

The problem was considered as a binary classification task, where the main objective was to predict whether the customer would subscribe to a term deposit. The dependent variable indicated the subscription as yes or no. The central focus was to evaluate how different machine learning models including logistic regression, naïve bayes and random forest could learn from different patterns and then make predictions accurately.

3.1 Dataset Description

The dataset used in this study is the Bank Marketing dataset from the UCI Machine Learning Repository (UCI, 2020). This dataset contains information and data from a Portuguese bank's telemarketing campaigns. These campaigns were conducted with the aim of promoting term deposits and getting maximum subscribers. This Bank Marketing dataset is quite popular in the machine learning community as it is mostly used to determine how well different machine learning algorithms can predict about the behavior of the customers.

This dataset consists of 45,211 records collected between 2008 and 2013. It includes sixteen independent features and one dependent variable or target variable that indicates whether the customer will subscribe or not.

3.1.1 Independent and target variable (features)

This dataset contains the following independent variables .

Demographic data: Variables related to demographic data include age, job, marital status and education.

Financial data: Variables related to financial data include balance, housing loan and personal loan

Campaign related data: Campaign related data include contact type, month, day, duration, campaign, pdays, previous, poutcome.

Target variable

y (binary: yes/no). The target variable indicates whether client or customer subscribed to a term deposit or not.

The features include both categorical and numerical variables. Categorical variables include variables such as job, marital status, education, housing, and loan status and numerical variables include variables like age, balance, duration, campaign, and previous contacts. One numerical variable named Duration was treated very carefully as it might cause data leakage. This variable records the length of the phone call, and thus it could have a direct impact on the results of the model.

It became clear after the initial analysis that the dataset is not balanced, with only around 11.3 percent of customers subscribed to a term deposit. Such an imbalance in a dataset can have negative impacts on the results shown by the models. The models might up with the results that are in favor of only the majority class, which in this case are the customers who did not subscribe to the term deposit. It might result in higher or better overall accuracy, but models will perform poorly while detecting the customers who subscribed to the term deposit. Additionally, some of the features like age and balance were found to be right skewed so normalization techniques like min-max scaling were applied to deal with this problem and bring the data in more usable form.

3.2 Data Preprocessing

3.2.1 Data Cleaning

The project begins with careful inspection of the data and then performing data cleaning actions. The actions included were resolving problems like missing and unknown values.

The job variable in the dataset had the highest cardinality or in simple terms it had the most unique categories among different categorical fields. It included fields like values like admin, technician,

services, etc. Some of the variables especially those with the categorical variables included values marked as “unknown”. All “unknown” entries in categorical variables were then replaced with proper missing value indicators (NA). This was done with the sole purpose that R’s built in functions would treat these values as genuine missing data. Additionally, rows which contain any NA values were then removed or dropped using `na.omit()` function. This makes sure that only complete cases in the dataset remained for the analysis. A final check was then carried out to confirm that none of the missing values remain, resulting in a clean dataset before starting the data modeling process.

3.2.2 Handling Class Imbalance

One of the main issues that arose early was that of class imbalance as most of the data belonged to customers who did not subscribe to term deposit. This issue was resolved using a technique called synthetic oversampling, used with the package called ROSE package in R, this package and technique help to balance the dataset and enhance the predictions for the minority class (Lemaître, Nogueira and Aridas, 2017; Chawla et al., 2002).

3.2.3 Feature Engineering and Transformation

After the data cleaning step, the next step involved data preparation so that it can be used in model training. Categorical data for example, job or marital status of the customers, was converted into numbers using technique called one hot encoding. This technique is very useful for models like Logistic regression and Naïve bayes. These numerical values were also scaled and transformed (normalization) so that it can help in better performance of the models. Outliers can negatively impact the result and accuracy of the models; therefore, outliers were spotted with the help of visualization tools and statistical methods which employ boxplots and Inter-quartile range (IQR) method respectively. They were then adjusted or capped so that they would no longer have much effect on the performance of the models (James et al., 2013).

3.3 Exploratory Data Analysis (EDA)

For the better understanding of the dataset, an exploratory data analysis was done using two packages in R which included ggplot2 and corrplot. This step was extremely beneficial as it helped in finding unusual patterns, it also helped to identify which features are related to each other and determine which features are most important in terms of making final predictions about the term deposit.

During the phase of exploratory data analysis (EDA), a package called ggplot2 was used quite often to create visualizations such as boxplots, confusion matrix and histograms and bar charts. This visualization helped a lot, especially to clearly understand the relationship between different features and the target or independent variable. For example, it was observed that the customers or clients with higher average balance and those who were contacted via cell phones were more interested in subscribing to the term deposit. In addition to it, the previous marketing outcome feature called poutcome indicated a strong relation with the success of the marketing campaign. This is because the customers who had shown or given positive response in the past were more likely to subscribe again (Moro, Cortez and Rita, 2014).

Different correlation graphs clearly suggested that features such as call duration, account balance, job type and the result of previous job marketing campaigns had the strongest impact on the possibility of subscription of term deposit (Moro, Cortez and Rita, 2014).

For numerical variables a correlation matrix was then computed using a package called corrplot. This highlighted different associations for variables such as “pdays” which represent the number of days since last contact and “previous” which represent the number of previous contacts. These variables were very important in terms of both the predictions and the interpretability. Therefore, these variables were retained in the dataset.

3.4 Model Selection

For this project three machine learning models were selected . These models include Logistic Regression, Naïve Bayes, and Random Forest and these models were selected based on different important factors. Firstly, interpretability was considered as a crucial factor because the models were required to be easy and simple to understand, especially to show which features are more important to predict if a customer will subscribe to a term deposit. Secondly, performance was considered as the other important factor because the model needed to perform better with the dataset that contains both the numeric and categoric types of data. Thirdly, it was important to explore different types of models. Logistic regression is a simple linear model, Naïve Bayes uses probabilities to make predictions, and Random Forest is an ensemble model that combines many decision trees. The last reason for selecting these models is that all three models are good for binary classification problems, which matches the aim of this project, predicting yes or no for term deposit subscription.

3.4.1 Logistic Regression

Logistic regression was selected for this project because it is well suited for binary classification problems, like to predict whether a customer will subscribe to a term deposit or not. It gives clear results which ultimately helps to understand which features are more important. It performs well when the data is linearly separable and is very efficient in terms of computation which makes it suitable for large datasets like the bank marketing dataset.

3.4.2 Naïve Bayes

Naïve Bayes was selected due to multiple reasons. It is known to be a simple model that is fast and easy to train and understand. It works well with datasets that contain a lot of categorical variables. It does not require a large amount of training data to give good results, and it can also deal with missing values in a better way. All these factors were considered before selecting naïve bayes model.

3.4.3 Random Forest

Random Forest was model included because it is very effective at dealing with complex data where features may have non-linear relationships. This model can avoid overfitting, it can automatically

select important features that can affect predictions. The most important reason to select the random forest model was that it gives good results even if the data is imbalanced. Thus, it is very effective with real world data and for real world problems.

3.5 Model Training and Evaluation

After the exploratory data analysis (EDA) the dataset was divided into training and testing sets using an 80/20 split. 10-fold cross-validation was performed on the training set to make sure that the data leakage does not occur, and the models did not overfit. The balanced dataset that was generated after applying the ROSE package was used for the purpose of training the machine learning models. In the end, the evaluation of the models was carried out on the original unbalanced testing set to ensure that the performance metrics reflected realistic conditions.

After the completion of exploratory data analysis, the next phase involved the training of three machine learning models on balanced and cleaned dataset. Testing of these three models was done using 10-fold cross-validation to make sure the results were more reliable. Another package called caret was used for the fine tuning of the hyperparameters with the help of grid search functionality. To ensure that the comparison is fair, all three models were evaluated using the performance metrics like accuracy, precision, recall, F1-score, and area under the curve (AUC). Among these metrics, area under the curve (AUC) was extremely helpful because it reflected better idea of model performance without depending upon a particular decision threshold (James et al., 2013).

The evaluation metrics suggest that out of three models, the model that gave best result was random forest. The random forest model was very good at dealing with lot of features from the data and its ability to automatically figure out the important features was excellent (Breiman, 2001). Logistic regression model was also very useful as it provided valuable insights into how different features affected the prediction. Apart from it, logistic regression model was very easy to understand and interpret. Naïve Bayes model, although it was fast and quite simple and easy to understand, it did not perform as well as the other two models, especially when it came to the evaluation metric recall, which indicated that it failed to identify positive cases where customers subscribed to the term deposit.

3.6 Summary of Methodology

In summary, the approach or methodology used in this project was based on the idea to keep a balance between computational speed and accuracy and make it easy to understand and interpret the results. It ensured that all the phases of machine learning process which include quality of data, handling the missing values, resolving the issue of class imbalance, training of the three models and evaluation of the model performance were carefully and thoroughly done. This helped provide important insight into behavior of the customer in the bank marketing. The bank marketing dataset included a diverse range of features and real-world challenges, this makes this dataset suitable for applying and implementing different supervised machine learning techniques for predicting the behavior of the customers.

4 Results and Discussion

4.1 Model Evaluation

The performance of the three models was evaluated using various metrics including accuracy, precision, recall, F1 score, and AUC. Additionally, confusion matrices were plotted along with box and scatter plots.

4.1.1 Naive Bayes model Evaluation

Evaluation of the Naive Bayes model depicts that it achieved an accuracy of approximately 77.36% with precision around 77.30%, recall of 77.48% and FI score of 77.4%. In addition to it, its AUC was 0.83, which indicates a good distinguishing ability or power.

4.1.1.1 Confusion matrix-Naïve Bayes

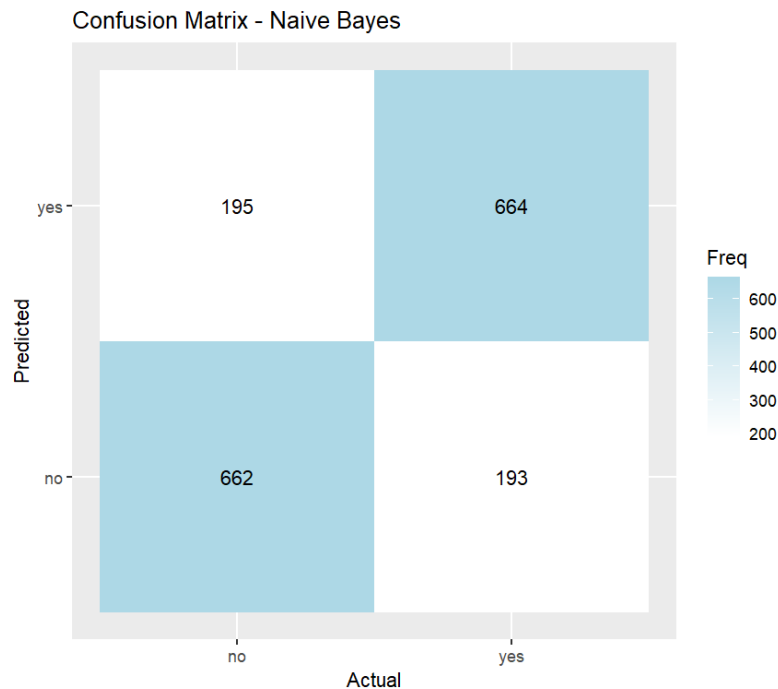


Figure 1 Confusion matrix of naive bayes model

4.1.2 Logistic regression model Evaluation

Evaluation of the Logistic Regression clearly demonstrated slightly improved performance with an accuracy of 77.54%, precision and recall of 79.35% and 74.45% respectively, FI score of 76.82% and an AUC of 0.84. This performance clearly indicates that the model is good at discriminating between different classes, however its recall was slightly lower than that of Naive Bayes.

4.1.2.1 Confusion Matrix- Logistic Regression

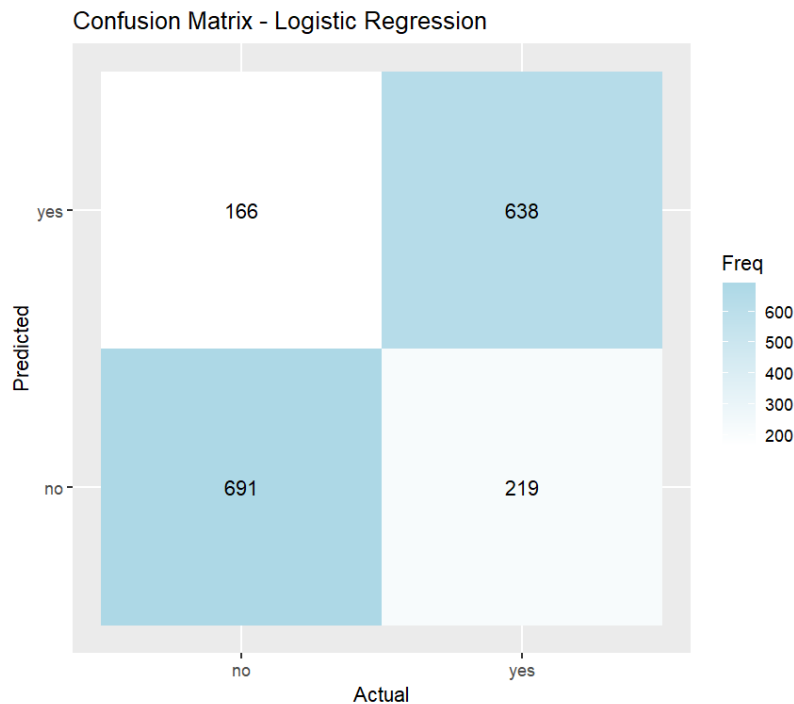


Figure 2 Confusion matrix of logistic regression model

4.1.3 Random Forest model Evaluation

Evaluation of the Random Forest model clearly indicates that it outperformed the other two models, achieving an accuracy of 92.18%, precision of 88.09%, recall of 97.55%, and an F1 score of 92.58%. Also, area under the curve (AUC) of Random Forest model was extremely high at 0.99, which clearly demonstrates its superior ability to classify subscription outcomes correctly.

4.1.3.1 Confusion Matrix-Random Forest

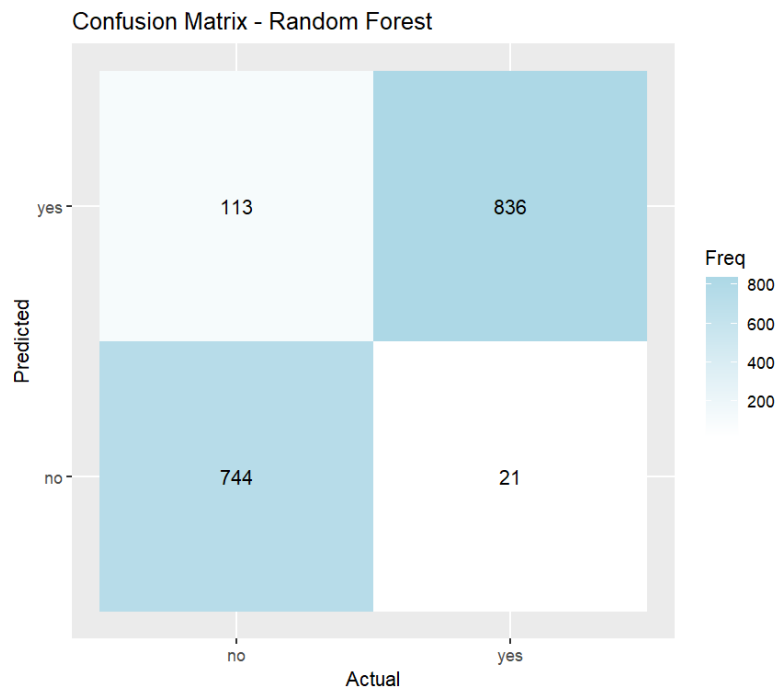


Figure 3 Confusion matrix of random forest model.

4.1.4 Metrics Comparison

The table below depicts a summary and comparison of the model performances.

Model	Accuracy	Precision	Recall	F1	AUC
Naive Bayes	0.7736289	0.7729919	0.7747958	0.7738928	0.8300100
Logistic Regression	0.7753792	0.7935323	0.7444574	0.7682119	0.8437264
Random Forest	0.9218203	0.8809273	0.9754959	0.9258029	0.9852740

These metrics clearly indicate that the two models Naive Bayes and Logistic Regression provide a reasonable baseline while Random Forest model delivers significantly improved performance in terms of both predictive accuracy and overall classification ability.

4.1.5 Age Distribution by Subscription Status

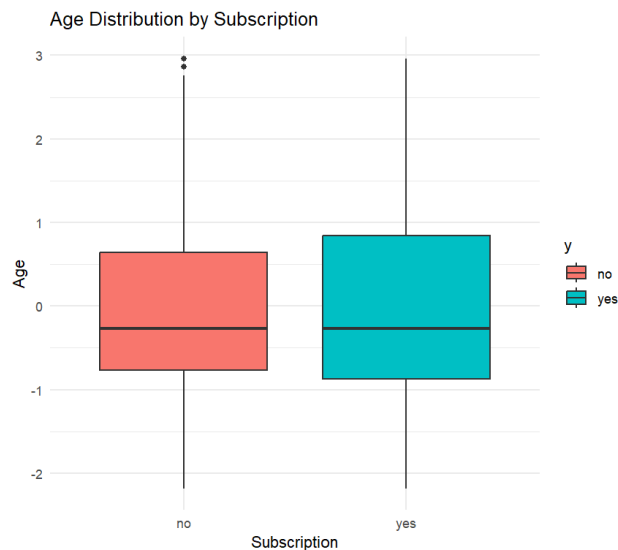


Figure 4 Age Distribution by Subscription

Figure 4.1 depicts the age distribution across customers who subscribed to term deposits and those who did not subscribe. The boxplot shows that the age distributions between subscribers and non-subscribers are somehow similar as both groups show similar median values. This tells us that age cannot be considered a strong differentiator for term deposit subscription behavior of customers. There were outliers in the upper range from both the groups, which suggests that most customers are from similar age range, however, there are some older individuals in both categories.

4.1.6 Relationship Between Age and Account Balance

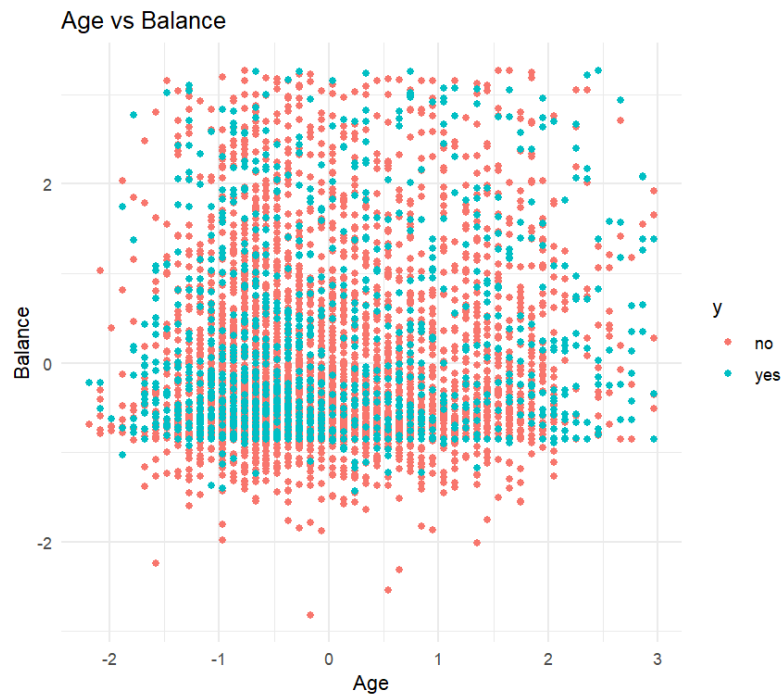


Figure 5 Age vs Balance by Subscription Status

Figure 4 depicts the relationship between the age of the customer and account balance, colored by the status of subscription. The scatter plot depicts a widely spread distribution of both subscribers and non-subscribers across different age and balance combinations. The separation between the two classes is not clear which suggests that the relationship between age, balance, and subscription behavior is complex and likely involves interactions with other variables. The data points for both subscription outcomes appear to be evenly distributed, though there might be a higher number of "yes" responses in different parts of the plot. The plot also reveals some outliers with very low balance values, mostly among non-subscribers.

4.1.7 ROC Curve Analysis

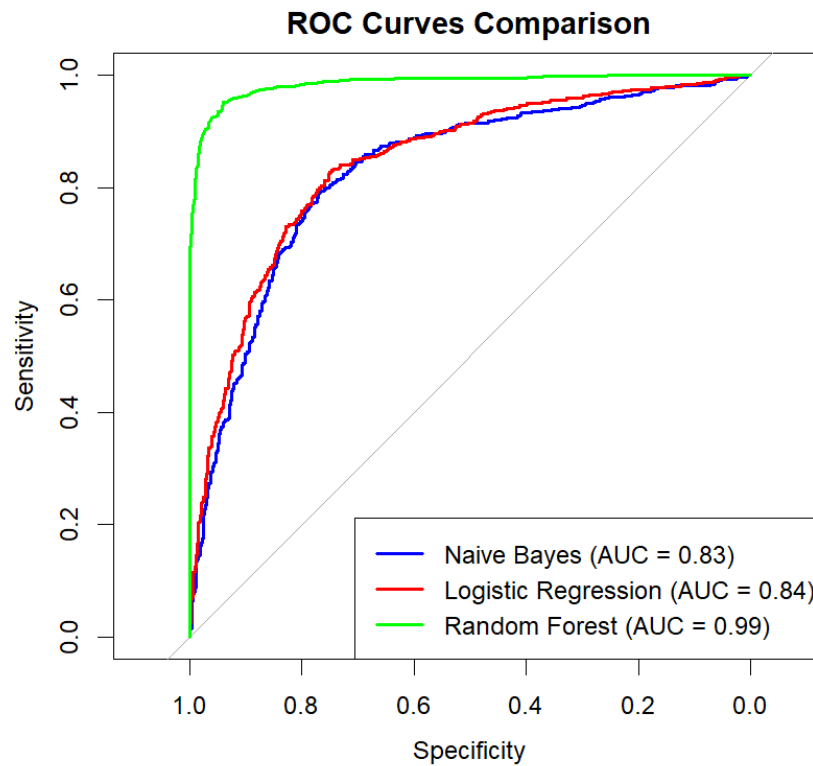


Figure 6 ROC Curves Comparison

Figure 6 displays the ROC curves for all three models, it demonstrates their ability to distinguish between classes. The Random Forest model represented by the green line, $AUC = 0.99$, dramatically outperforms both Naive Bayes model represented by blue line, $AUC = 0.83$ and Logistic Regression model represented by red line, $AUC = 0.84$. The curve of a random forest model climbs rapidly towards the top-left corner. It indicates a very good sensitivity (true positive rate) even at high specificity levels. This means the model can identify subscribers effectively while minimizing false positives. On the other hand, Naive Bayes and Logistic Regression models have almost similar overall AUC values, their curves intersect at several points which suggest that each model performs better in different threshold regions.

4.1.8 Feature Importance

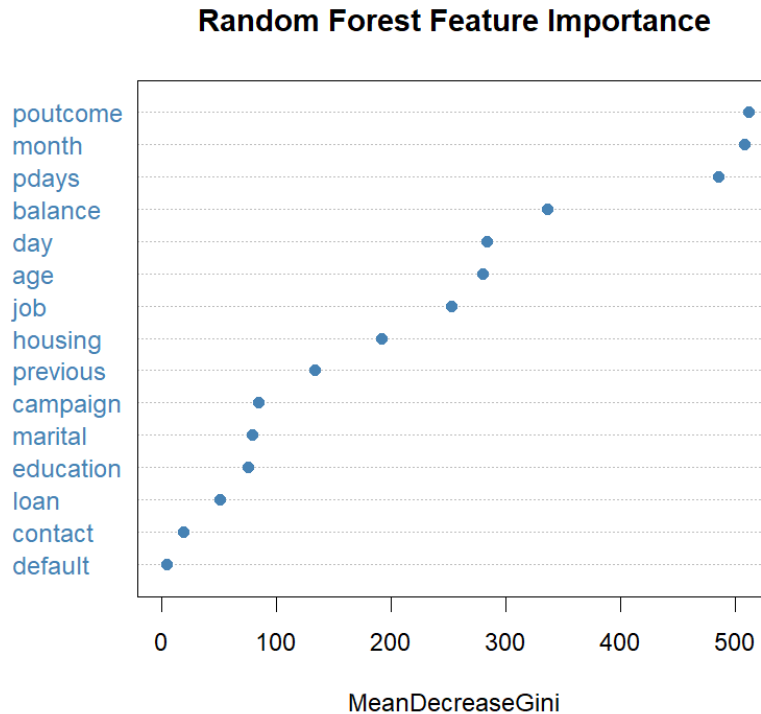


Figure 7 Random Forest Feature Importance

Figure 7 displays the importance of features in the Random Forest model which is measured by mean decrease in Gini impurity. The analysis of feature importance suggests that there are three features that have the highest influence to predict the term deposit subscription . These features are the previous marketing outcome (poutcome), contact month (month), and days since last contact (pdays). The timing and results of previous marketing contacts are crucial factors to determine the likelihood of subscription. The group of the second most important features includes account balance, contact day, and customer age. The analysis also reveals that demographic factors such as education, marital status, and job type have less influence than campaign-related variables. The least important features include default status and contact type (contact).

5 Conclusion

This study clearly demonstrates the importance of data preprocessing, exploratory data analysis (EDA) and advanced predictive modeling using machine learning techniques for introducing effective strategies for bank marketing. The research is organized in clear sections like data preparation, exploratory data analysis(EDA), followed by testing and evaluation of the models and experimental results. Therefore, it clearly provides a transparent view of the entire process. After detail evaluation of the three models including Naïve Bayes, Logistics Regression and Random Forest, it is concluded that the Random Forest model emerged as the most effective technique, achieving an accuracy of 92.18%, precision of 88.09%, recall of 97.55%, F1-score of 92.58%, and AUC of 0.985. In comparison, Logistic Regression achieved 77.53% accuracy and Naïve Bayes achieved 77.36 %. These findings highlight the importance of different data driven methods or approaches in improving customer engagement in the banking sector. By implementing the insights and methods used in this study, banks can not only improve the efficiency of their marketing campaigns but also enhance customer experience by matching the preference and need of the customers with the product offers. The goal and objective are not just the effective and better marketing approach but making more personalized customer relationships and building stronger connections that will ultimately benefit both the banks and their customers.

6 Appendix

1. Bank-full.csv



bank-full.csv

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